

MFE5200 – Computational Methods in Financial Engineering
Summer Term, 2025-26

Assignment 1 (Solution)

Problem 1

$$\begin{aligned} E[\hat{\mu}_1] &= \frac{1}{2n} \sum_{i=1}^{2n} [E[h(X_i)] + E[f(X_i)]] \\ &= \frac{1}{2n} 2n [E[h(X)] + E[f(X)]] \\ &= E[h(X)] + E[f(X)] \\ &= E[h(X) + f(X)]. \end{aligned}$$

$$\begin{aligned} E[\hat{\mu}_2] &= \frac{1}{n} \sum_{i=1}^n E[h(X_i)] + \frac{1}{n} \sum_{i=1}^n E[f(Y_i)] \\ &= E[h(X)] + E[f(X)] \\ &= E[h(X) + f(X)]. \end{aligned}$$

So, both $\hat{\mu}_1$ and $\hat{\mu}_2$ are unbiased estimators for μ .

$$\begin{aligned} \text{var}[\hat{\mu}_1] &= \frac{1}{4n^2} \sum_{i=1}^{2n} \text{var}[h(X_i) + f(X_i)] \\ &= \frac{1}{4n^2} (2n) \text{var}(h(X) + f(X)) \\ &= \frac{1}{2n} [\text{var}(h(X)) + 2\text{cov}(h(X), f(X)) + \text{var}(f(X))] \\ \text{var}[\hat{\mu}_2] &= \frac{1}{n^2} \sum_{i=1}^n \text{var}[h(X_i)] + \frac{1}{n^2} \sum_{i=1}^n \text{var}[f(Y_i)] \\ &= \frac{1}{n} \text{var}[h(X)] + \frac{1}{n} \text{var}[f(X)] \end{aligned}$$

$$\begin{aligned}
& \text{var}[\hat{\mu}_1] - \text{var}[\hat{\mu}_2] \\
&= \frac{1}{2n} \left[\text{var}(h(X)) + 2\text{cov}(h(X), f(X)) + \text{var}(f(X)) \right] - \frac{1}{n} \text{var}[h(X)] - \frac{1}{n} \text{var}[f(X)] \\
&= -\frac{1}{2n} \left[\text{var}(h(X)) - 2\text{cov}(h(X), f(X)) + \text{var}(f(X)) \right] \\
&= -\frac{1}{2n} \text{var}(f(X) - h(X)) \leq 0.
\end{aligned}$$

Hence, $\text{var}(\hat{\mu}_1) \leq \text{var}(\hat{\mu}_2)$.

Problem 2

(a)

By Itô's lemma, we have

$$\begin{aligned}
dY(t) &= \frac{\partial Y(t)}{\partial t} dt + \frac{\partial Y(t)}{\partial S(t)} dS(t) + \frac{1}{2} \frac{\partial^2 Y(t)}{\partial S^2(t)} (dS(t))^2 \\
&= \alpha S^{\alpha-1}(t) dS(t) + \frac{1}{2} \alpha(\alpha-1) S^{\alpha-2}(t) (dS(t))^2 \\
&= \alpha S^\alpha(t) (rdt + \sigma dW(t)) + \frac{1}{2} \alpha(\alpha-1) S^\alpha(t) (rdt + \sigma dW(t))^2 \\
&= S^\alpha(t) \left(\left(\alpha r + \frac{1}{2} \alpha(\alpha-1) \sigma^2 \right) dt + \alpha \sigma dW(t) \right) \\
&= Y(t) \left(\left(\alpha r + \frac{1}{2} \alpha(\alpha-1) \sigma^2 \right) dt + \alpha \sigma dW(t) \right).
\end{aligned}$$

$$Y(0) = S^2.$$

(b)

By Monte-Carlo method, we have

$$\hat{V}_n = \frac{\sum_{k=1}^n V_k}{n},$$

where

$$V_k = \max \left(\frac{\sum_{i=1}^m Y^{(k)}(t_i)}{m} - K, 0 \right) \text{ and } \{Y^{(k)}(t_1), Y^{(k)}(t_2), \dots, Y^{(k)}(t_m)\} \text{ is the } k\text{-th path}$$

of $Y(t)$.

Method 1 (Biased Estimator)

By discretizing the SDE of $Y(t)$ in (a), we have

$$Y(t_i) - Y(t_{i-1}) \approx \left(\alpha r + \frac{1}{2} \alpha (\alpha - 1) \sigma^2 \right) Y(t_{i-1}) (t_i - t_{i-1}) + \sigma Y(t_{i-1}) (W(t_i) - W(t_{i-1}))$$
$$Y(t_i) \approx Y(t_{i-1}) + \left(\alpha r + \frac{1}{2} \alpha (\alpha - 1) \sigma^2 \right) Y(t_{i-1}) (t_i - t_{i-1}) + \sigma \sqrt{t_i - t_{i-1}} Y(t_{i-1}) Z, \quad Z \sim N(0, 1).$$

So, $\{Y^{(k)}(t_1), Y^{(k)}(t_2), \dots, Y^{(k)}(t_m)\}$ can be obtained as follows

$$Y^{(k)}(t_i) \approx Y^{(k)}(t_{i-1}) + \left(\alpha r + \frac{1}{2} \alpha (\alpha - 1) \sigma^2 \right) Y^{(k)}(t_{i-1}) (t_i - t_{i-1}) + \sigma \sqrt{t_i - t_{i-1}} Y^{(k)}(t_{i-1}) Z.$$

Method 2 (Unbiased Estimator)

By solving the SDE of $Y(t)$ in (a), we have

$$Y(t_i) = Y(t_{i-1}) \exp \left(\left(\alpha r + \frac{1}{2} \alpha (\alpha - 1) \sigma^2 - \frac{1}{2} \alpha^2 \sigma^2 \right) (t_i - t_{i-1}) + \alpha \sigma (W(t_i) - W(t_{i-1})) \right)$$
$$= Y(t_{i-1}) \exp \left(\left(\alpha r - \frac{1}{2} \alpha \sigma^2 \right) (t_i - t_{i-1}) + \alpha \sigma \sqrt{t_i - t_{i-1}} Z \right).$$

So, $\{Y^{(k)}(t_1), Y^{(k)}(t_2), \dots, Y^{(k)}(t_m)\}$ can be obtained as follows

$$Y^{(k)}(t_i) = Y^{(k)}(t_{i-1}) \exp \left(\left(\alpha r - \frac{1}{2} \alpha \sigma^2 \right) (t_i - t_{i-1}) + \alpha \sigma \sqrt{t_i - t_{i-1}} Z \right).$$

Problem 3

As $U \in [0, 1]$, then $10U(1 - U) \in [0, 2.5]$.

Since $X = \text{int}(10U(1 - U))$, X can have 3 possible values which are 0, 1 and 2.

$$\begin{aligned} P(X = 0) &= P(10U(1 - U) < 1) \\ &= P(10U^2 - 10U + 1 > 0) \\ &= P\left(0 < U < \frac{1}{2} - \frac{\sqrt{60}}{20}\right) + P\left(\frac{1}{2} + \frac{\sqrt{60}}{20} < U < 1\right) \\ &= \frac{1}{2} - \frac{\sqrt{60}}{20} + \left(1 - \frac{1}{2} - \frac{\sqrt{60}}{20}\right) \\ &= 1 - \sqrt{\frac{3}{5}} = 0.2254. \end{aligned}$$

$$\begin{aligned}
P(X=1) &= P(1 \leq 10U(1-U) < 2) \\
&= P(10U(1-U) < 2) - P(10U(1-U) < 1) \\
&= P(10U^2 - 10U + 2 > 0) - \left(1 - \sqrt{\frac{3}{5}}\right) \\
&= P\left(0 < U < \frac{1}{2} - \frac{1}{\sqrt{20}}\right) + P\left(\frac{1}{2} + \frac{1}{\sqrt{20}} < U < 1\right) - \left(1 - \sqrt{\frac{3}{5}}\right) \\
&= \frac{1}{2} - \frac{1}{\sqrt{20}} + \left(1 - \frac{1}{2} - \frac{1}{\sqrt{20}}\right) - \left(1 - \sqrt{\frac{3}{5}}\right) \\
&= 1 - \frac{2}{\sqrt{20}} - 1 + \sqrt{\frac{3}{5}} \\
&= \sqrt{\frac{3}{5}} - \sqrt{\frac{1}{5}} = 0.3274
\end{aligned}$$

$$\begin{aligned}
P(X=2) &= P(2 \leq 10U(1-U)) \\
&= P(10U(1-U) \geq 2) \\
&= P\left(\frac{1}{2} - \frac{1}{\sqrt{20}} \leq U \leq \frac{1}{2} + \frac{1}{\sqrt{20}}\right) \\
&= \frac{2}{\sqrt{20}} \\
&= \frac{1}{\sqrt{5}} = 0.4472.
\end{aligned}$$

Problem 4

Since the sample space of X is $\{0, 1, 2, 3, 4, 5\}$, the sample space of Y is $\{0, 1, 4, 9\}$.

$$P(Y = 0) = P(X = 3) = \binom{5}{3} (0.5)^5 = \frac{10}{32},$$

$$P(Y = 1) = P(X = 2 \text{ or } X = 4) = \binom{5}{2} (0.5)^5 + \binom{5}{4} (0.5)^5 = \frac{15}{32},$$

$$P(Y = 4) = P(X = 1 \text{ or } X = 5) = \binom{5}{1} (0.5)^5 + \binom{5}{5} (0.5)^5 = \frac{6}{32},$$

$$P(Y = 9) = P(X = 0) = \binom{5}{0} (0.5)^5 = \frac{1}{32}.$$

Problem 5

Uniform number u	Simulated value of x
$0 \leq u < 0.1$	$0.1x = u \Rightarrow x = \frac{u}{0.1}.$
$0.1 \leq u \leq 0.3$	$x = 1.$
$0.3 < u \leq 0.6$	$x = 3.$
$0.6 < u \leq 1$	$0.2x = u \Rightarrow x = \frac{u}{0.2}.$

Problem 6

Let

$$h(x) = \frac{f(x)}{g(x)} = \frac{\frac{1}{2}x^2e^{-x}}{\lambda e^{-\lambda x}} = \frac{1}{2\lambda}x^2e^{-x(1-\lambda)}.$$

Consider

$$h'(x) = \frac{dh(x)}{dx} = \frac{1}{2\lambda} \left(2xe^{-x(1-\lambda)} - x^2(1-\lambda)e^{-x(1-\lambda)} \right).$$

Setting $h'(x) = 0$, we have

$$\frac{1}{2\lambda} \left(2xe^{-x(1-\lambda)} - x^2(1-\lambda)e^{-x(1-\lambda)} \right) = 0$$

$$x(2 - x(1-\lambda)) = 0$$

$$x = \frac{2}{1-\lambda}. \quad (\text{since } x > 0)$$

It is easy to check that

$$h'(x) > 0 \quad \text{when } x < \frac{2}{1-\lambda};$$

$$h'(x) < 0 \quad \text{when } x > \frac{2}{1-\lambda}.$$

So,

$$\frac{f(x)}{g(x)} \leq h\left(\frac{2}{1-\lambda}\right) = \frac{2e^{-2}}{\lambda(1-\lambda)^2}, \quad \text{for } x > 0.$$

Under the acceptance-rejection method, the expected number of trial samples needed to generate a sample from f is given by

$$h\left(\frac{2}{1-\lambda}\right) = \frac{2e^{-2}}{\lambda(1-\lambda)^2}.$$

Define

$$k(\lambda) = \frac{2e^{-2}}{\lambda(1-\lambda)^2}.$$

Consider

$$k'(\lambda) = \frac{-2e^{-2}((1-\lambda)^2 - 2\lambda(1-\lambda))}{\lambda^2(1-\lambda)^4} = \frac{-2e^{-2}(1-3\lambda)}{\lambda^2(1-\lambda)^3}.$$

Setting $k'(x) = 0$, we have

$$\frac{-2e^{-2}(1-3\lambda)}{\lambda^2(1-\lambda)^3} = 0$$

$$\lambda = \frac{1}{3}.$$

By the first derivative test, $k(x)$ has a minimum at $\lambda = \frac{1}{3}$ and

$$k\left(\frac{1}{3}\right) = \frac{2e^{-2}}{\left(\frac{1}{3}\right)\left(1-\frac{1}{3}\right)^2} = 1.827.$$

Problem 7

It is clear that $\Pr(|X| \leq x) = 0$ when $x \leq 0$.

Suppose $x > 0$, consider

$$\begin{aligned}\Pr(|X| \leq x) &= \Pr(-x \leq X \leq x) \\ &= \Pr(X \leq x) - \Pr(X \leq -x) \\ &= \int_{-\infty}^x \frac{1}{2} e^{-|s|} ds - \int_{-\infty}^{-x} \frac{1}{2} e^{-|s|} ds \\ &= \int_{-\infty}^0 \frac{1}{2} e^s ds + \int_0^x \frac{1}{2} e^{-s} ds - \int_{-\infty}^{-x} \frac{1}{2} e^s ds \\ &= \frac{1}{2} - \frac{1}{2}(e^{-x} - 1) - \frac{1}{2} e^{-x} \\ &= 1 - e^{-x}.\end{aligned}$$

So, the pdf of $|X|$ is given by

$$\begin{aligned}f(x) &= \frac{d \Pr(|X| \leq x)}{dx} \\ &= \begin{cases} 0 & \text{if } x \leq 0 \\ e^{-x} & \text{if } x > 0. \end{cases}\end{aligned}$$

Hence, $|X|$ has the exponential distribution with mean 1.

Problem 8

Let $g(x, y)$ be the joint pdf of X and Y , let $f(s, \theta)$ be the joint pdf of S and Θ .

Since Θ is uniform on $[0, 2\pi]$, S is an exponential random variable with mean of 2, and Θ and S are independent, we have

$$f(s, \theta) = \left(\frac{1}{2} e^{-\frac{s}{2}} \right) \left(\frac{1}{2\pi} \right) = \frac{1}{4\pi} e^{-\frac{s}{2}}.$$

You are given that

$$x = \sqrt{s} \cos \theta \quad \text{and} \quad y = \sqrt{s} \sin \theta.$$

From the hint, we have

$$f(s, \theta) = g(x, y) |J|$$

where

$$\begin{aligned}
|J| &= \begin{vmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial \theta} \end{vmatrix} \\
&= \begin{vmatrix} \frac{1}{2\sqrt{s}} \cos \theta & -\sqrt{s} \sin \theta \\ \frac{1}{2\sqrt{s}} \sin \theta & \sqrt{s} \cos \theta \end{vmatrix} \\
&= \frac{1}{2} (\cos^2 \theta + \sin^2 \theta) = \frac{1}{2}.
\end{aligned}$$

So,

$$\begin{aligned}
g(x, y) &= \frac{1}{|J|} f(s, \theta) \\
&= 2 \left(\frac{1}{4\pi} e^{-\frac{s}{2}} \right) \\
&= \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}} \\
&= \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \right) \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \right).
\end{aligned}$$

Therefore, X and Y are independent standard normal random variables

Problem 9

Let

$$A = \begin{pmatrix} A_{11} & 0 & 0 \\ A_{21} & A_{22} & 0 \\ A_{31} & A_{32} & A_{33} \end{pmatrix}.$$

We have

$$\begin{pmatrix} A_{11} & 0 & 0 \\ A_{21} & A_{22} & 0 \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ 0 & A_{22} & A_{32} \\ 0 & 0 & A_{33} \end{pmatrix} = \begin{pmatrix} 3.21 & 1.82 & 2.18 \\ 1.82 & 5.89 & 3.23 \\ 2.18 & 3.23 & 8.12 \end{pmatrix}$$

$$A_{11}^2 = 3.21 \Rightarrow A_{11} = \sqrt{3.21} = 1.7916.$$

$$A_{21}A_{11} = 1.82 \Rightarrow A_{21} = \frac{1.82}{1.7916} = 1.0159.$$

$$A_{31}A_{11} = 2.18 \Rightarrow A_{31} = \frac{2.18}{1.7916} = 1.2168.$$

$$A_{21}^2 + A_{22}^2 = 5.89 \Rightarrow A_{22}^2 = 5.89 - (1.0159)^2 = 4.8579 \Rightarrow A_{22} = 2.2041.$$

$$A_{31}A_{21} + A_{32}A_{22} = 3.23 \Rightarrow (1.2168)(1.0159) + 2.2041 \times A_{32} = 3.23 \Rightarrow A_{32} = 0.9046.$$

$$A_{31}^2 + A_{32}^2 + A_{33}^2 = 8.12 \Rightarrow (1.2168)^2 + (0.9046)^2 + A_{33}^2 = 8.12 \Rightarrow A_{33} = 2.4127.$$

So,

$$A = \begin{pmatrix} 1.7916 & 0 & 0 \\ 1.0159 & 2.2041 & 0 \\ 1.2168 & 0.9046 & 2.4127 \end{pmatrix}.$$

Problem 10

(a)

Use (ii) in Hint by taking $g(x) = x^2$, $h(x) = x$ and $k(x) \equiv 1$, we have

$$\left(E[X|Y]\right)^2 = E\left[\left(E[X|Y]\right)^2 \middle| Y\right].$$

Use (ii) in Hint again by taking $g(x) \equiv h(x) \equiv k(x) \equiv 1$, $(E[X|Y])^2$ can also be expressed as

$$\left(E[X|Y]\right)^2 = E[X|Y]E[X|Y] = E\left[E[X|Y]X \middle| Y\right].$$

So,

$$\begin{aligned} \text{Var}(X|Y) &= E[X^2|Y] - 2\left(E[X|Y]\right)^2 + \left(E[X|Y]\right)^2 \\ &= E[X^2|Y] - 2E\left[E[X|Y]X \middle| Y\right] + E\left[\left(E[X|Y]\right)^2 \middle| Y\right] \\ &= E\left[X^2 - 2E[X|Y]X + \left(E[X|Y]\right)^2 \middle| Y\right] \\ &= E\left[\left(X - E[X|Y]\right)^2 \middle| Y\right] \\ &\geq 0. \end{aligned}$$

(b)

$$\begin{aligned}E[Var[X|Y]] &= E\left[E[X^2|Y] - (E[X|Y])^2\right] \\&= E\left[E[X^2|Y]\right] - E\left[(E[X|Y])^2\right] \quad (\text{use (i) in Hint}) \\&= E[X^2] - E\left[(E[X|Y])^2\right].\end{aligned}$$

$$\begin{aligned}Var[E[X|Y]] &= E\left[(E[X|Y])^2\right] - (E[E[X|Y]])^2 \\&= E\left[(E[X|Y])^2\right] - (E[X])^2. \quad (\text{use (i) in Hint})\end{aligned}$$

Hence,

$$\begin{aligned}&E[Var[X|Y]] + Var[E[X|Y]] \\&= E[X^2] - E\left[(E[X|Y])^2\right] + E\left[(E[X|Y])^2\right] - (E[X])^2 \\&= E[X^2] - (E[X])^2 \\&= Var[X].\end{aligned}$$